

# Transport & Logistics Management System (TMS)

Comprehensive End-to-End System Verification & Real-Time Sync Audit Report

REPORT DATE

28-09-2026

PHASES STATUS

22 / 22 (100%)

UNIT TESTS

68 / 68 Passed

EXCEL & LIVE SYNC

100% Verified

**Executive Summary:** The Transport & Logistics Management System (TMS) has undergone end-to-end integration testing, data persistence audits, real-time Google Sheets synchronization checks, and production build verification. All 22 designated phases are 100% complete and fully verified.

## 1. Transport Enquiry Records Audit (Created & Verified)

| ID        | Route (Origin → Destination)     | Type     | Vehicle No | Container No | Freight (₹) | Status      |
|-----------|----------------------------------|----------|------------|--------------|-------------|-------------|
| ENQ-10001 | Mumbai Port → Bhiwandi Warehouse | Import   | MH04JK9876 | MSCU1234567  | ₹45,000     | Verified    |
| ENQ-10002 | Bengaluru ICD → Chennai Port     | Export   | KA01AB1234 | CMAU9876543  | ₹62,000     | Verified    |
| ENQ-10003 | Hazira Plant → Pithampur Area    | Empty    | GJ06CD5678 | TLLU5678901  | ₹88,000     | Verified    |
| ENQ-10004 | Gurugram Hub → Jaipur Park       | Offload  | DL01EF9012 | HAPU3456789  | ₹38,000     | Verified    |
| ENQ-10005 | Sriperumbudur → Hyderabad        | Flatrack | TN09GH3456 | COSU7890123  | ₹75,000     | Verified    |
| ENQ-10017 | Chennai Port → Sriperumbudur Hub | Import   | TN05AB9988 | TGHU7654321  | ₹55,000     | Live Synced |

## 2. Page-by-Page & Real-Time Sync Verification

| Application Page | API Endpoint         | Live Sync Mechanism                             | HTTP Status |
|------------------|----------------------|-------------------------------------------------|-------------|
| View Enquiry     | /api/enquiries       | Dual-sheet atomic write (enquiries + movements) | 200 OK      |
| Pending Bills    | /api/billing/pending | Real-time query of completed enquiries          | 200 OK      |

| Application Page | API Endpoint              | Live Sync Mechanism                               | HTTP Status |
|------------------|---------------------------|---------------------------------------------------|-------------|
| Processed Bills  | /api/billing/bills        | Atomic FY bill sequence under LockService         | 200 OK      |
| Daily Report     | /api/reports/daily        | Automated syncDailyRow trigger on save            | 200 OK      |
| Billing Report   | /api/reports/billing      | Live calculation over processed_bills & payments  | 200 OK      |
| Vendor Reports   | /api/vendors/report       | Live calculation over enquiries & vendor_payments | 200 OK      |
| Ageing Analysis  | /api/billing/ageing       | 4-Bucket receivables ageing (0-30d to 90+d)       | 200 OK      |
| Excel Exporter   | /api/exports/master-excel | Full 20-sheet .xlsx live workbook link            | 200 OK      |

### 3. Excel & Google Sheets Live Sync Architecture

- **Daily Report Workbook:** 23 standard columns (Creation Date, Booking Date, Company, Client, Vehicle, Container, Movement In/Out, Advances, Comments) synced in real time.
- **Client-Wise Report Workbook:** Dynamic dedicated tab per client with instant upserts on enquiry creation/update.
- **Processed Bills Workbook:** 12 columns tracking Bill Number, Financial Year, Subtotal, Tax, Total, Paid Amount, and Pending Balance.
- **Master Excel Snapshot:** Live 20-tab .xlsx workbook export available on demand.

### 4. Test Harness & Build Verification

- **Vitest Unit Testing:** 68 out of 68 unit tests passing across 13 test suites.
- **Next.js Production Build:** 58 static and dynamic routes compiled cleanly with zero errors.
- **Security Hardening:** Single-user salted SHA-256 password digest, session proxy cookies, and 60-second timeout budget.